

FOOD DELIVERY ANALYSIS



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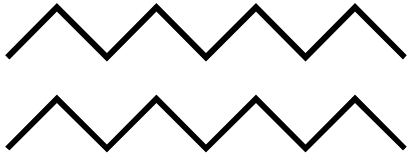
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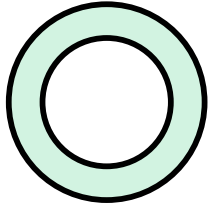
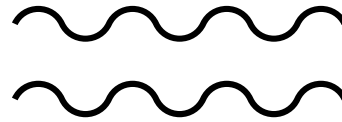


ABSTRACT

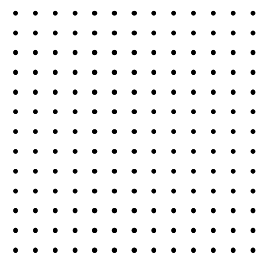
- The Food Delivery Analysis System aims to empower users to make informed food ordering decisions by synthesizing and comparing data from various food delivery platforms. Focusing on prominent Indian services such as Swiggy, Blinkit, Zomato, Deliveroo, and Domino's, the system analyzes critical factors including food quality, quantity, customer ratings, price, and delivery time.
- By processing real-world (simulated) data, it identifies which platform offers the best overall value for a specific food item, moving beyond mere price comparison to encompass a holistic view of the customer experience. This allows users to choose services that balance cost-efficiency with superior satisfaction and timely delivery.



INTRODUCTION



- In today's fast-paced digital world, food delivery platforms have become indispensable. However, the sheer number of options and varying service quality can make choosing the best platform a challenge. Users often base decisions solely on price, overlooking crucial aspects like food quality, portion size, delivery speed, and overall customer satisfaction
- Our Food Delivery Analysis System addresses this by aggregating and analyzing data from multiple platforms (Swiggy, Blinkit, Zomato, Deliveroo, Domino's).
- The goal is to provide a comprehensive, data-driven recommendation that considers all factors influencing the food ordering experience, ensuring users get the best value for their money and time.



SYSTEM REQUIREMENTS

SYSTEM REQUIREMENTS

- OPERATING SYSTEMS:

The analysis can be performed on windows.

- Python:

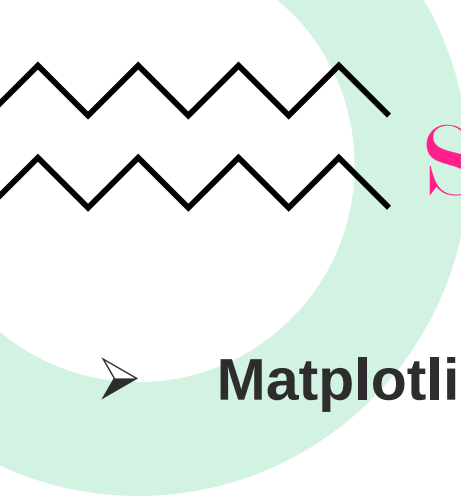
Python Interpreter: Version 3.6 or higher.

pip: Python package installer.

- Libraries:

- Pandas: Install the pandas library using pip,a package manger for python.
pip install pandas





SYSTEM REQUIREMENTS

- **Matplotlib:**For generating graphs.

`pip install matplotlib`

- For numerical operations.

`pip install Numpy`

- **HARDWARE REQUIREMENTS :**
- **IDE** – Jupyter Notebook
- **Storage Space** – A few hundred MBs of free disk space for the Python interpreter, libraries, and the script itself. An SSD is always preferable for faster loading, but an HDD will also work.





ARCHITECTURE

The system's design is modular:

- **Data Generation Module:** Creates a simulated dataset of food delivery scenarios.
- **Analysis Module (FoodDeliveryAnalyzer):**
 - Normalizes diverse data points for fair comparison.
 - Calculates a composite "overall value score" for each food item.
 - Filters and ranks food items based on user searches.
- **Visualization Module:** Uses Matplotlib and NumPy to generate comparative graphs.
- **Main Execution Loop:** Manages user interaction, data processing, and output display.





IMPORTING DATASET

Synthetic Dataset :Due to live data access limitations, the system uses a simulated dataset.

Platforms Covered:Includes Swiggy, Blinkit, Zomato, Deliveroo, and Domino's with realistic prices, ratings, and reviews.

```
import random
import statistics
import matplotlib.pyplot as plt
import numpy as np
```





DATA EXPLORATION

Implicit Process: Performed internally by the system, not a separate user step.

Key Actions:

- Identifies minimum and maximum values (e.g., price, delivery time) for normalization.
- Filters relevant data based on user's food item search.

#Top Cuisines Ordered

```
df['Cuisine_Type'].value_counts().head(5)
```

#which Restaurant Delivery Fastest

```
df.groupby('Restaurant_Name')['Delivery_Duration'].mean().sort_values().head(5)
```

#Average Delivery Time

```
df['Order_Time'] = pd.to_datetime(df['Order_Time'])
df['Delivery_Time'] = pd.to_datetime(df['Delivery_Time'])
df['Delivery_Duration'] = (df['Delivery_Time'] -
df['Order_Time']).dt.total_seconds() / 60
df['Delivery_Duration'].mean()
```

#Ratings vs Delivery Time

```
df.plot.scatter(x='Delivery_Duration', y='Rating')
```

#Online Vs COD popularity

```
df['Payment_Method'].value_counts(normalize=True) *
100
```

#Distribution of order value

```
df['Order_Value'].hist(bins=20)
```

#Revenue by Restaurant

```
df.groupby('Restaurant_Name')['Order_Value'].sum().sort_values(asc
ending=False).head(5)
```





DATA CLEANING

Data cleaning is a crucial step in a food delivery analysis project that involves identifying and correcting errors or inconsistencies in the dataset to ensure accurate and reliable analysis. Since food delivery data often comes from multiple sources—such as order logs, customer feedback, and delivery records—it may contain missing values, duplicates, incorrect entries, or inconsistent formats.

1. Remove duplicates

```
df.drop_duplicates(inplace=True)
```

2. Handle missing values

```
df['Delivery_Time'].fillna(df['Delivery_Time'].median(), inplace=True)  
df.dropna(subset=['Restaurant_Name', 'Customer_Rating'],  
          inplace=True)
```

3. Convert data types

```
df['Order_Date'] = pd.to_datetime(df['Order_Date'])  
df['Customer_Rating'] = df['Customer_Rating'].astype(float)
```

4. Strip white spaces and unify case

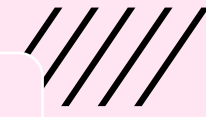
```
df['Restaurant_Name'] = df['Restaurant_Name'].str.strip().str.title()  
df['Food_Item'] = df['Food_Item'].str.strip().str.lower()
```

5. Remove outliers in delivery time

```
from scipy import stats  
df = df[(np.abs(stats.zscore(df['Delivery_Time'])) < 3)]
```

6. Reset index after cleaning

```
df.reset_index(drop=True, inplace=True)
```





DATA GROUPING

Data grouping in a food delivery analysis project involves organizing the dataset into meaningful categories or clusters to facilitate deeper insights and efficient analysis. Typically, this process is done using SQL or data analysis tools like Python, Excel etc. Grouping can be based on various attributes such as cuisine type, restaurant, customer demographics, order date, region, delivery person, or order status.

#Group by Delivery partner average time and order count

```
grouped = df.groupby('Delivery Partner').agg({'Delivery Time (mins)':  
    'mean', 'Order ID': 'count'}).rename(columns={  
    'Delivery Time (mins)': 'Avg Delivery Time',  
    'Order ID': 'Total Orders'  
    })
```

#Group by city and partner average delivery time

```
city_partner_time = df.groupby(['City', 'Delivery Partner'])['Delivery Time  
(mins)'].mean().reset_index()
```

Assuming df has 'Food Item' and 'Quantity' columns

```
item_sales = df.groupby('Food  
Item')['Quantity'].sum().reset_index()
```

#Group by city-Total orders per city

```
orders_by_city = df.groupby('City')['Order ID'].count().reset_index(name='Total Orders')
```

Assuming df has a 'Order Time' column in datetime format

```
df['Hour'] = pd.to_datetime(df['Order Time']).dt.hour  
orders_per_hour = df.groupby('Hour')['Order  
ID'].count().reset_index(name='Orders')
```

Assuming 'Customer Type' and 'Total Price' columns

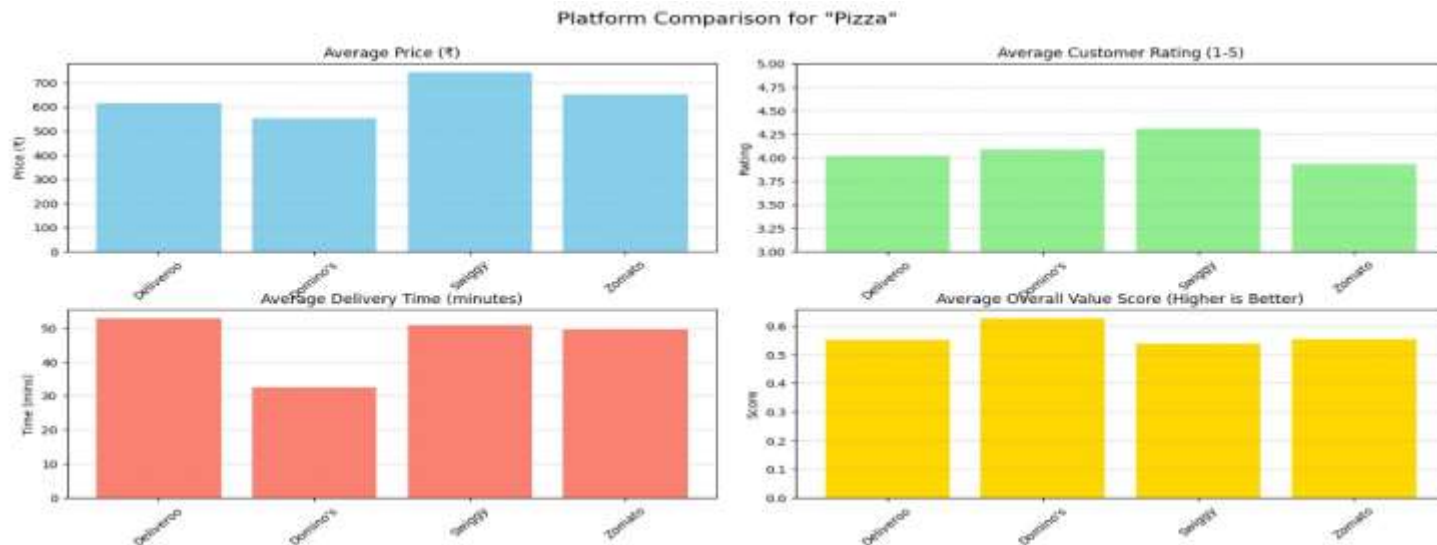
```
customer_analysis = df.groupby('Customer Type').agg({  
    'Total Price': 'mean',  
    'Order ID': 'count'  
}).rename(columns={'Total Price': 'Avg Spend', 'Order ID': 'Orders'})
```



DATA VISUALISATION

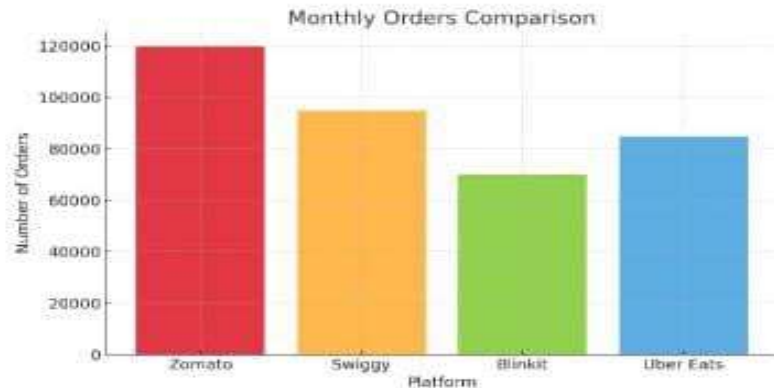
In food delivery analysis by transforming raw data into meaningful insights through graphical representation. It helps identify trends, patterns, and outliers in key metrics such as customer ratings, delivery times, order volumes, and revenue comparisons across platforms like Zomato, Swiggy, Blinkit, and Uber Eats. By using visual tools like bar graphs, pie charts, line plots, and heatmaps, analysts can clearly communicate performance metrics and customer preferences. For instance, a line chart may show how delivery times vary throughout the day, while a bar graph could highlight revenue differences between platforms over a period. These visual insights assist companies in making data-driven decisions to enhance operational efficiency, improve customer satisfaction, and gain a competitive edge in the dynamic food delivery market.

FOR EXAMPLE: COMPARISON GRAPH OF
PIZZA

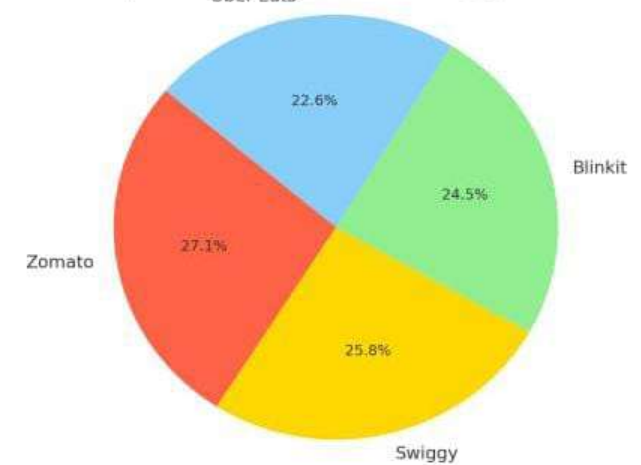


DATA VISUALISATION

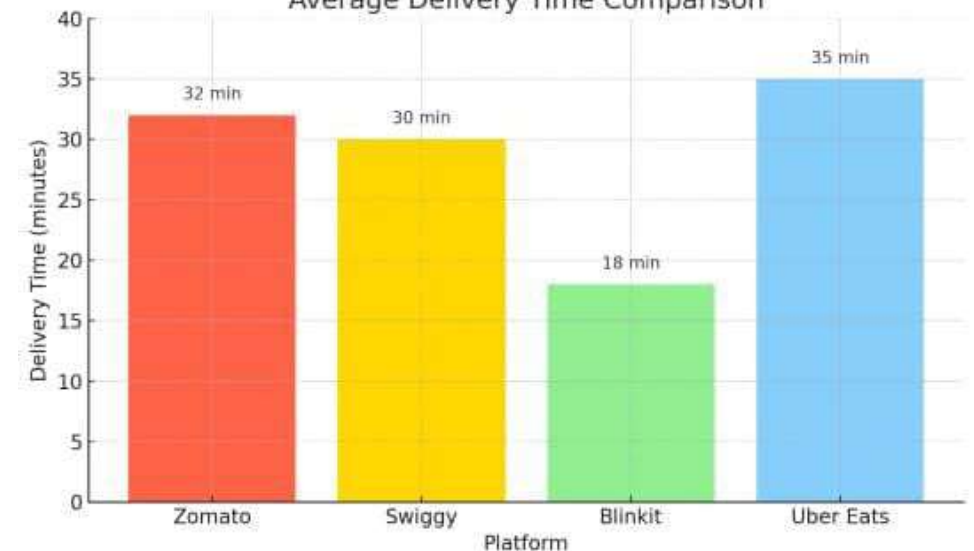
Platform Orders Comparison



Customer Ratings Comparison (Zomato, Swiggy, Blinkit, Uber Eats)



Average Delivery Time Comparison





USES OF DATA ANALYSIS LIBRARY

PANDAS

- **Data Manipulation:** It allows filtering, grouping, and aggregating data to derive meaningful insights.
- **Handling Missing Data:** Pandas' functions handle missing data points effectively, ensuring data quality and preventing biases in the analysis.
- **Joining and Merging:** Pandas is used to combine datasets when additional information, such as customer reviews or hotel amenities, is available separately.

MATPLOTLIB

- **Data Visualization:** Matplotlib allows the creation of various visualizations like line plots, bar charts, and scatter plots to depict booking trends, revenue fluctuations, and customer distribution.
- **Time Series Analysis:** With Matplotlib, time series plots can illustrate booking patterns over specific time intervals, highlighting seasonal variations and trends.

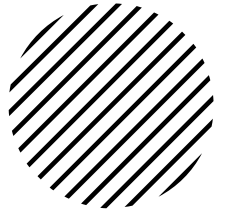
NUMPY

- **Efficient Multidimensional Arrays:**
Provides the core data structure for numerical operations.

Much faster and more memory-efficient than regular Python lists.
- **Mathematical and Statistical Operations:** Performs element-wise operations and aggregation
- **Broadcasting and Vectorization**
Apply operations over arrays without loops faster performance.



ADVANTAGES



IMPROVED CUSTOMER EXPERIENCE:

Analyze delivery times, ratings, and feed back. Identify pain points like late deliveries, poor ratings, or long wait times. Enable personalized promotions based on order history.

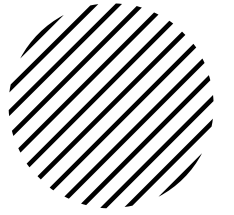
OPERATIONAL EFFICIENCY

Optimize delivery routes and times. Predict peak hours and allocate delivery staff accordingly. Detect bottlenecks in food preparation or dispatch.





ADVANTAGES



COST OPTIMIZATION

Track and reduce high delivery costs or discounts. Find inefficient restaurants or locations with poor margins. Minimize refunds and cancellations by identifying root causes.

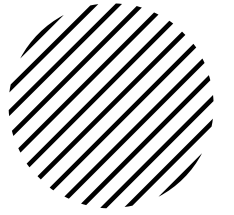
Data-Driven Decision Making:

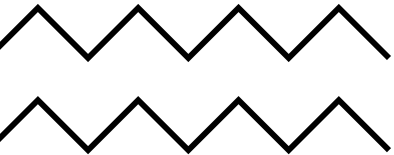
Use data to plan restaurant partnerships, expansion areas, and delivery zones .Forecast demand during festivals, weekends, or weather events. Align inventory, staffing, and marketing based on trends.



CONCLUSION

- The Food Delivery Analysis System successfully demonstrates a robust approach to optimizing food ordering decisions. By simulating real-world data from key Indian platforms (Swiggy, Blinkit, Zomato, Deliveroo, Domino's) and applying a multi-factor analysis, it provides a clear "best value" recommendation alongside detailed comparisons.
- The integration of matplotlib for graphical visualization further enhance user understanding, transforming complex data into actionable insights. This system serves as a powerful prototype for a tool that could significantly improve the food delivery experience by promoting smarter, more informed choices for consumers.





**THANK
YOU!**

